

API Design

1. Authentication APIs:

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/auth/login	Authenticate user and generate JWT token	Email, Password	JWT token and role
POST	/api/auth/logout	Logout current user	Token	Success message
PUT	/api/auth/change-password	Update password	Old password, New password	Success message

2. User Management APIs (Admin)

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/users	Create student or teacher account	Name, Email, Role	User created message
GET	/api/users	Retrieve all users	None	List of users
PUT	/api/users/:id	Update user information	Name, Email, Role	Updated message
DELETE	/api/users/:id	Delete user account	User ID	Success message

3. Course Management APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/courses	Create course	Course code, Course name, Credit	Course created message
GET	/api/courses	Retrieve all courses	None	List of courses
PUT	/api/courses/:id	Update course information	Course details	Updated message
DELETE	/api/courses/:id	Delete course	Course ID	Success message

4. Assignment APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/assignments	Create assignment	Title, Description, Deadline	Assignment created message
GET	/api/assignments	Retrieve all assignments	None	Assignment list
GET	/api/assignments/:id	Retrieve a specific assignment	Assignment ID	Assignment details
PUT	/api/assignments/:id	Update assignment	Updated assignment information	Updated message
DELETE	/api/assignments/:id	Delete assignment	Assignment ID	Success message

5. CT Notice APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/ct-notices	Publish CT notice	Title, Exam date	CT notice published message
GET	/api/ct-notices	View CT notices	None	List of CT notices

6. Course Materials APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/materials	Upload course material	Title, File	Upload success message
GET	/api/materials	Retrieve all materials	None	Material list
GET	/api/materials/:id	Download or view material	Material ID	Material details/file

7. Routine APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/routines	Create class routine	Day, Time, Course information	Routine created message
GET	/api/routines	Retrieve class routines	None	Routine list
PUT	/api/routines/:id	Update routine	Updated routine information	Updated message

8. Exam Routine APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/exam-routines	Create exam schedule	Course, Date, Time	Schedule created message
GET	/api/exam-routines	Retrieve exam schedules	None	Exam routine list
PUT	/api/exam-routines/:id	Update exam schedule	Updated schedule information	Updated message

9. Notice APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/notices	Publish academic notice	Title, Description	Notice published message
GET	/api/notices	Retrieve all notices	None	Notice list
DELETE	/api/notices/:id	Remove notice	Notice ID	Success message

10. Dashboard APIs

Method	Endpoint	Purpose	Input Data	Expected Output
GET	/api/dashboard/student	Load student dashboard data	None	Upcoming classes, exams and assignments

11. AI APIs

Method	Endpoint	Purpose	Input Data	Expected Output
POST	/api/ai/chat	Answer academic questions using AI	Question	AI-generated answer
POST	/api/ai/study-plan	Generate a personalized study plan	Subject, Number of days	Study plan
POST	/api/ai/quiz	Generate quizzes based on a topic	Topic	List of questions
POST	/api/ai/summarize	Summarize lecture notes or text content	Long text content	Short summary
POST	/api/ai/assignment-helper	Provide guidance and assistance for assignments	Assignment query	Explanation and recommendations

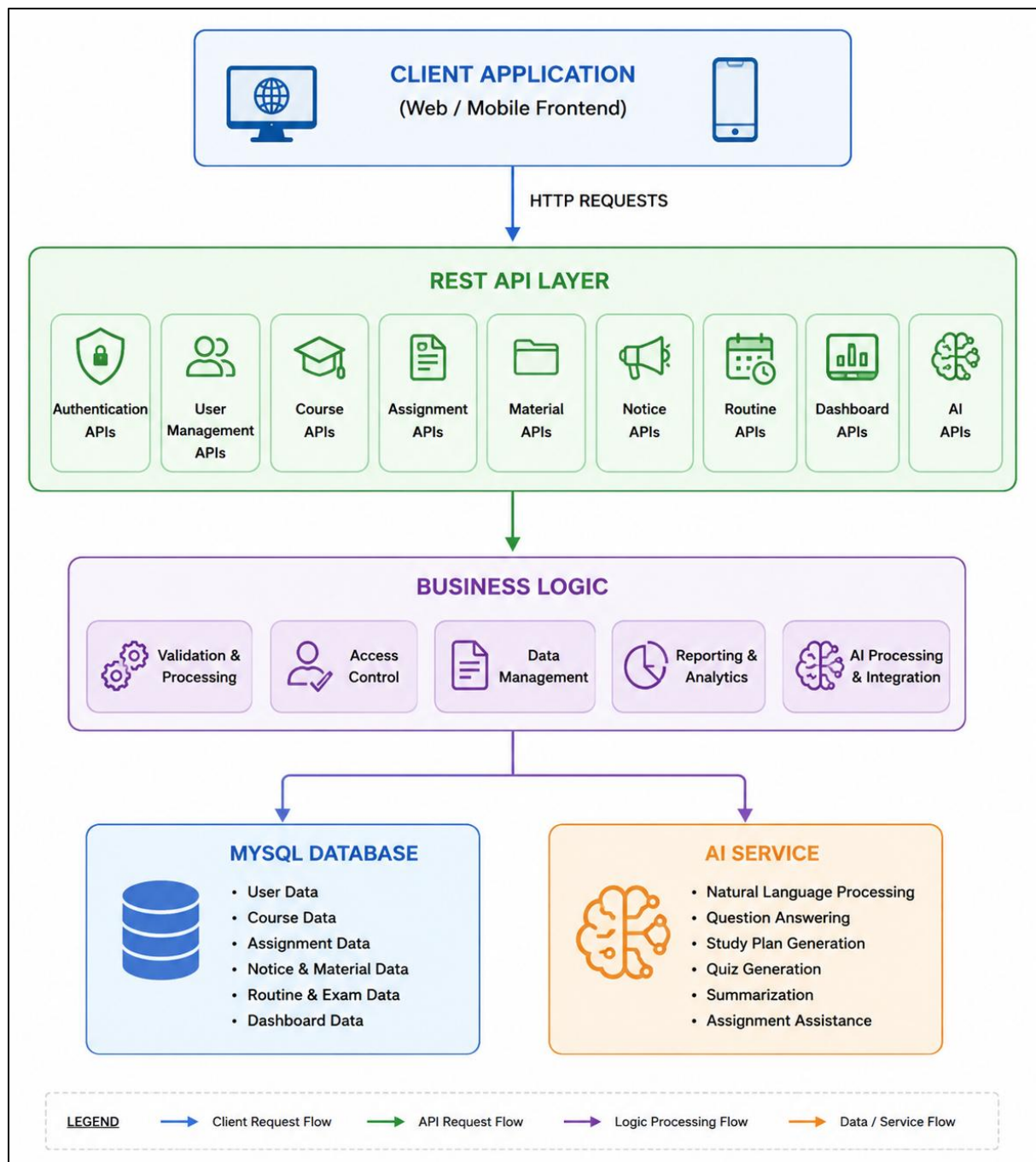


Fig : API Architecture Diagram